

Migrate PSP DB to AWS RDS with Goldengate

- Scope:
- 1. Pre-requisites
 - 1.1 Create RDS
 - 1.2 Create Goldengate Hub
- 2. setup replication from primary to parallel environment
 - 2.1 Start GoldenGate extract and pump from Primary to parallel environment
 - 2.2 start replicate
- 3. export backup on parallel environment
 - 3.1 Nullify plain text columns on target
 - 3.2 start export backup PSPADM schema by excluding IBOB tables
- 4. setup replication from primary to AWS
 - 4.1 Create Goldengate Hub
 - 4.2 Setup the migration environment on Goldengate Hub
 - 4.3 Import data to the target RDS db
- 4.4 Final step for Goldengate replication from IHP to AWS
- 4.5 Setup Veridata
- 4.6 Setup Goldengate replication from AWS to IHP
- 4.7 Setup RDS and GG monitoring

Scope:

POC on setting up GG replication by excluding plain text columns on primary from extract/source site.

1. Pre-requisites

- Goldengate version 12.3 software install on source and target.
- Make sure OGG mountpoint has space available to retain 30 days trail files.

1.1 Create RDS

- Create AWS credential file on your MAC or PC after you create your access key through AWS console

```
For Prod, you should add [sbg-qbo-prod] section to ~/.aws/credentials
```

```
[sbg-qbo-prod]
aws_access_key_id=<aws access key id>
aws_secret_access_key=<aws secret access key>
region=us-west-2
```

```
For PreProd, you should add [sbg-qbo-ppd] section to ~/.aws/credentials
```

```
[sbg-psp-ppd]
aws_access_key_id=<aws access key id>
aws_secret_access_key=<aws secret access key>
region=us-west-2
```

```
wdhl0753fb3a0:~$ chmod 700 ~/.aws/credentials
```

- Download RDS creation code from Github <https://github.com/payroll-services-platform/desktop-payroll-database-engineering/tree/master/aws-rds> to your Mac or PC. **Please always redownload the code each db migration because it is keeping change for new feature or bug fix.**
- Cd to the directory aws-rds-master. You should have the following files

```
ls
README.md                                create_rds_dependency_w_cli.sh
create_rds_w_cf.sh                       cf_rds_template.json
```

- Execute script to create RDS-dependency resources. Output would look like below

```
banl141bbfff3:prod kpatat$ ./create_rds_dependency_w_cli.sh prod clusterdb vpc-2
{
  "DBParameterGroup": {
    "DBParameterGroupName": "oracle-ee-12-1-psp-prod-clusterdb-vpc-2",
    "DBParameterGroupFamily": "oracle-ee-12.1",
    "Description": "oracle-ee-12-1-psp-prod-clusterdb-vpc-2",
    "DBParameterGroupArn": "arn:aws:rds:us-west-2:893547637742:pg:oracle-ee-12-1-psp-prod-clusterdb-vpc-2"
  }
}
{
  "DBParameterGroupName": "oracle-ee-12-1-psp-prod-clusterdb-vpc-2"
}
{
  "DBParameterGroupName": "oracle-ee-12-1-psp-prod-clusterdb-vpc-2"
}
{
  "DBParameterGroupName": "oracle-ee-12-1-psp-prod-clusterdb-vpc-2"
}
{
  "DBParameterGroupName": "oracle-ee-12-1-psp-prod-clusterdb-vpc-2"
}
{
  "DBParameterGroupName": "oracle-ee-12-1-psp-prod-clusterdb-vpc-2"
}
{
  "DBParameterGroupName": "oracle-ee-12-1-psp-prod-clusterdb-vpc-2"
}
{
  "DBParameterGroupName": "oracle-ee-12-1-psp-prod-clusterdb-vpc-2"
}
{
  "DBParameterGroupName": "oracle-ee-12-1-psp-prod-clusterdb-vpc-2"
}
{
  "DBParameterGroupName": "oracle-ee-12-1-psp-prod-clusterdb-vpc-2"
}
{
  "DBParameterGroupName": "oracle-ee-12-1-psp-prod-clusterdb-vpc-2"
}
{
  "DBParameterGroupName": "oracle-ee-12-1-psp-prod-clusterdb-vpc-2"
}
{
  "OptionGroup": {
    "OptionGroupName": "oracle-ee-12-1-prod-clusterdb-vpc-2",
    "OptionGroupDescription": "oracle-ee-12-1-prod-clusterdb-vpc-2",
    "EngineName": "oracle-ee",
    "MajorEngineVersion": "12.1",
    "Options": [],
    "AllowsVpcAndNonVpcInstanceMemberships": true,
    "OptionGroupArn": "arn:aws:rds:us-west-2:893547637742:og:oracle-ee-12-1-prod-clusterdb-vpc-2"
  }
}
{
  "OptionGroup": {
    "OptionGroupName": "oracle-ee-12-1-prod-clusterdb-vpc-2",
    "OptionGroupDescription": "oracle-ee-12-1-prod-clusterdb-vpc-2",
    "EngineName": "oracle-ee",
    "MajorEngineVersion": "12.1",
    "Options": [
      {
        "OptionName": "Timezone",
        "OptionDescription": "Change time zone",
        "Persistent": true,
        "Permanent": false,
        "OptionSettings": [
          {
            "Name": "TIME_ZONE",
            "Value": "US/Pacific",
            "DefaultValue": "UTC",
            "Description": "Specifies the timezone the user wants to change the system time to",
            "ApplyType": "DYNAMIC",
```

```

        "DataType": "STRING",
        "AllowedValues": "Africa/Cairo,Africa/Casablanca,Africa/Harare,Africa/Lagos,Africa/Luanda,Africa/Monrovia,Africa/Nairobi,Africa/Tripoli,Africa/Windhoek,America/Araguaina,America/Argentina/Buenos_Aires,America/Asuncion,America/Bogota,America/Caracas,America/Chicago,America/Chihuahua,America/Cuiaba,America/Denver,America/Detroit,America/Fortaleza,America/Godthab,America/Guatemala,America/Halifax,America/Lima,America/Los_Angeles,America/Manaus,America/Matamoros,America/Mexico_City,America/Monterrey,America/Montevideo,America/New_York,America/Phoenix,America/Santiago,America/Sao_Paulo,America/Tijuana,America/Toronto,Asia/Amman,Asia/Ashgabat,Asia/Baghdad,Asia/Baku,Asia/Bangkok,Asia/Beirut,Asia/Calcutta,Asia/Damascus,Asia/Dhaka,Asia/Hong_Kong,Asia/Irkutsk,Asia/Jakarta,Asia/Jerusalem,Asia/Kabul,Asia/Karachi,Asia/Kathmandu,Asia/Kolkata,Asia/Krasnoyarsk,Asia/Magadan,Asia/Manila,Asia/Muscat,Asia/Novosibirsk,Asia/Rangoon,Asia/Riyadh,Asia/Seoul,Asia/Shanghai,Asia/Singapore,Asia/Taipei,Asia/Tehran,Asia/Tokyo,Asia/Ulaanbaatar,Asia/Vladivostok,Asia/Yakutsk,Asia/Yerevan,Atlantic/Azores,Atlantic/Cape_Verde,Australia/Adelaide,Australia/Brisbane,Australia/Darwin,Australia/Eucla,Australia/Hobart,Australia/Lord_Howe,Australia/Perth,Australia/Sydney,Brazil/DeNoronha,Brazil/East,Canada/Newfoundland,Canada/Saskatchewan,Etc/GMT-3,Europe/Amsterdam,Europe/Athens,Europe/Berlin,Europe/Dublin,Europe/Helsinki,Europe/Kaliningrad,Europe/London,Europe/Madrid,Europe/Moscow,Europe/Paris,Europe/Prague,Europe/Rome,Europe/Sarajevo,Pacific/Apia,Pacific/Auckland,Pacific/Chatham,Pacific/Fiji,Pacific/Guam,Pacific/Honolulu,Pacific/Kiritimati,Pacific/Marquesas,Pacific/Samoa,Pacific/Tongatapu,Pacific/Wake,US/Alaska,US/Central,US/East-Indiana,US/Eastern,US/Pacific,UTC",
        "IsModifiable": true,
        "IsCollection": false
    }
},
    "DBSecurityGroupMemberships": [],
    "VpcSecurityGroupMemberships": []
}
},
    "AllowsVpcAndNonVpcInstanceMemberships": false,
    "OptionGroupArn": "arn:aws:rds:us-west-2:893547637742:og:oracle-ee-12-1-prod-clusterdb-vpc-2"
}
}
{
    "OptionGroup": {
        "OptionGroupName": "oracle-ee-12-1-prod-clusterdb-vpc-2",
        "OptionGroupDescription": "oracle-ee-12-1-prod-clusterdb-vpc-2",
        "EngineName": "oracle-ee",
        "MajorEngineVersion": "12.1",
        "Options": [
            {
                "OptionName": "NATIVE_NETWORK_ENCRYPTION",
                "OptionDescription": "Oracle Advanced Security - Native Network Encryption",
                "Persistent": false,
                "Permanent": false,
                "OptionSettings": [
                    {
                        "Name": "SQLNET.CRYPTO_CHECKSUM_SERVER",
                        "Value": "REQUESTED",
                        "DefaultValue": "REQUESTED",
                        "Description": "Specifies the desired data integrity behavior",
                        "ApplyType": "STATIC",
                        "DataType": "STRING",
                        "AllowedValues": "ACCEPTED,REJECTED,REQUESTED,REQUIRED",
                        "IsModifiable": true,
                        "IsCollection": false
                    },
                    {
                        "Name": "SQLNET.ENCRYPTION_TYPES_SERVER",
                        "Value": "RC4_256,AES256,AES192,3DES168,RC4_128,AES128,3DES112,RC4_56,DES,RC4_40,DES40",
                        "DefaultValue": "RC4_256,AES256,AES192,3DES168,RC4_128,AES128,3DES112,RC4_56,DES,RC4_40,DES40",
                        "Description": "Specifies list of encryption algorithms in order of intended use",
                        "ApplyType": "STATIC",
                        "DataType": "STRING",
                        "AllowedValues": "RC4_256,AES256,AES192,3DES168,RC4_128,AES128,3DES112,RC4_56,DES,RC4_40,DES40",
                        "IsModifiable": true,
                        "IsCollection": true
                    }
                ],
                "Name": "SQLNET.CRYPTO_CHECKSUM_TYPES_SERVER",
                "Value": "SHA1,MD5",
                "DefaultValue": "SHA1,MD5",
            }
        ]
    }
}

```

```

        "Description": "Specifies list of checksumming algorithms in order of intended use",
        "ApplyType": "STATIC",
        "DataType": "STRING",
        "AllowedValues": "SHA1,MD5",
        "IsModifiable": true,
        "IsCollection": true
    },
    {
        "Name": "SQLNET.ENCRYPTION_SERVER",
        "Value": "REQUESTED",
        "DefaultValue": "REQUESTED",
        "Description": "Specifies the desired encryption behavior",
        "ApplyType": "STATIC",
        "DataType": "STRING",
        "AllowedValues": "ACCEPTED,REJECTED,REQUESTED,REQUIRED",
        "IsModifiable": true,
        "IsCollection": false
    }
],
"DBSecurityGroupMemberships": [],
"VpcSecurityGroupMemberships": []
},
{
    "OptionName": "Timezone",
    "OptionDescription": "Change time zone",
    "Persistent": true,
    "Permanent": false,
    "OptionSettings": [
        {
            "Name": "TIME_ZONE",
            "Value": "US/Pacific",
            "DefaultValue": "UTC",
            "Description": "Specifies the timezone the user wants to change the system time to",
            "ApplyType": "DYNAMIC",
            "DataType": "STRING",
            "AllowedValues": "Africa/Cairo,Africa/Casablanca,Africa/Harare,Africa/Lagos,Africa/Luanda,Africa/Monrovia,Africa/Nairobi,Africa/Tripoli,Africa/Windhoek,America/Araguaina,America/Argentina/Buenos_Aires,America/Asuncion,America/Bogota,America/Caracas,America/Chicago,America/Chihuahua,America/Cuiaba,America/Denver,America/Detroit,America/Fortaleza,America/Godthab,America/Guatemala,America/Halifax,America/Lima,America/Los_Angeles,America/Manaus,America/Matamoros,America/Mexico_City,America/Monterrey,America/Montevideo,America/New_York,America/Phoenix,America/Santiago,America/Sao_Paulo,America/Tijuana,America/Toronto,Asia/Amman,Asia/Ashgabat,Asia/Baghdad,Asia/Baku,Asia/Bangkok,Asia/Beirut,Asia/Calcutta,Asia/Damascus,Asia/Dhaka,Asia/Hong_Kong,Asia/Irkutsk,Asia/Jakarta,Asia/Jerusalem,Asia/Kabul,Asia/Karachi,Asia/Kathmandu,Asia/Kolkata,Asia/Krasnoyarsk,Asia/Magadan,Asia/Manila,Asia/Muscat,Asia/Novosibirsk,Asia/Rangoon,Asia/Riyadh,Asia/Seoul,Asia/Shanghai,Asia/Singapore,Asia/Taipei,Asia/Tehran,Asia/Tokyo,Asia/Ulaanbaatar,Asia/Vladivostok,Asia/Yakutsk,Asia/Yerevan,Atlantic/Azores,Atlantic/Cape_Verde,Australia/Adelaide,Australia/Brisbane,Australia/Darwin,Australia/Eucla,Australia/Hobart,Australia/Lord_Howe,Australia/Perth,Australia/Sydney,Brazil/DeNoronha,Brazil/East,Canada/Newfoundland,Canada/Saskatchewan,Etc/GMT-3,Europe/Amsterdam,Europe/Athens,Europe/Berlin,Europe/Dublin,Europe/Helsinki,Europe/Kaliningrad,Europe/London,Europe/Madrid,Europe/Moscow,Europe/Paris,Europe/Prague,Europe/Rome,Europe/Sarajevo,Pacific/Apia,Pacific/Auckland,Pacific/Chatham,Pacific/Fiji,Pacific/Guam,Pacific/Honolulu,Pacific/Kiritimati,Pacific/Marquesas,Pacific/Samoa,Pacific/Tongatapu,Pacific/Wake,US/Alaska,US/Central,US/East-Indiana,US/Eastern,US/Pacific,UTC",
            "IsModifiable": true,
            "IsCollection": false
        }
    ],
    "DBSecurityGroupMemberships": [],
    "VpcSecurityGroupMemberships": []
}
],
"AllowsVpcAndNonVpcInstanceMemberships": false,
"OptionGroupArn": "arn:aws:rds:us-west-2:893547637742:og:oracle-ee-12-1-prod-clusterdb-vpc-2"
}
}
{
    "OptionGroup": {
        "OptionGroupName": "oracle-ee-12-1-prod-clusterdb-vpc-2",
        "OptionGroupDescription": "oracle-ee-12-1-prod-clusterdb-vpc-2",
        "EngineName": "oracle-ee",
        "MajorEngineVersion": "12.1",
        "Options": [

```

```

{
  "OptionName": "NATIVE_NETWORK_ENCRYPTION",
  "OptionDescription": "Oracle Advanced Security - Native Network Encryption",
  "Persistent": false,
  "Permanent": false,
  "OptionSettings": [
    {
      "Name": "SQLNET.CRYPTO_CHECKSUM_SERVER",
      "Value": "REQUESTED",
      "DefaultValue": "REQUESTED",
      "Description": "Specifies the desired data integrity behavior",
      "ApplyType": "STATIC",
      "DataType": "STRING",
      "AllowedValues": "ACCEPTED,REJECTED,REQUESTED,REQUIRED",
      "IsModifiable": true,
      "IsCollection": false
    },
    {
      "Name": "SQLNET.ENCRYPTION_TYPES_SERVER",
      "Value": "RC4_256,AES256,AES192,3DES168,RC4_128,AES128,3DES112,RC4_56,DES,RC4_40,DES40",
      "DefaultValue": "RC4_256,AES256,AES192,3DES168,RC4_128,AES128,3DES112,RC4_56,DES,RC4_40,DES40",
      "Description": "Specifies list of encryption algorithms in order of intended use",
      "ApplyType": "STATIC",
      "DataType": "STRING",
      "AllowedValues": "RC4_256,AES256,AES192,3DES168,RC4_128,AES128,3DES112,RC4_56,DES,RC4_40,DES40",
      "IsModifiable": true,
      "IsCollection": true
    },
    {
      "Name": "SQLNET.CRYPTO_CHECKSUM_TYPES_SERVER",
      "Value": "SHA1,MD5",
      "DefaultValue": "SHA1,MD5",
      "Description": "Specifies list of checksumming algorithms in order of intended use",
      "ApplyType": "STATIC",
      "DataType": "STRING",
      "AllowedValues": "SHA1,MD5",
      "IsModifiable": true,
      "IsCollection": true
    },
    {
      "Name": "SQLNET.ENCRYPTION_SERVER",
      "Value": "REQUIRED",
      "DefaultValue": "REQUESTED",
      "Description": "Specifies the desired encryption behavior",
      "ApplyType": "STATIC",
      "DataType": "STRING",
      "AllowedValues": "ACCEPTED,REJECTED,REQUESTED,REQUIRED",
      "IsModifiable": true,
      "IsCollection": false
    }
  ],
  "DBSecurityGroupMemberships": [],
  "VpcSecurityGroupMemberships": []
},
{
  "OptionName": "Timezone",
  "OptionDescription": "Change time zone",
  "Persistent": true,
  "Permanent": false,
  "OptionSettings": [
    {
      "Name": "TIME_ZONE",
      "Value": "US/Pacific",
      "DefaultValue": "UTC",
      "Description": "Specifies the timezone the user wants to change the system time to",
      "ApplyType": "DYNAMIC",
      "DataType": "STRING",
      "AllowedValues": "Africa/Cairo,Africa/Casablanca,Africa/Harare,Africa/Lagos,Africa/Luanda,Africa/Monrovia,Africa/Nairobi,Africa/Tripoli,Africa/Windhoek,America/Araguaina,America/Argentina

```

```

/Buenos_Aires,America/Asuncion,America/Bogota,America/Caracas,America/Chicago,America/Chihuahua,America/Cuiaba,
America/Denver,America/Detroit,America/Fortaleza,America/Godthab,America/Guatemala,America/Halifax,America/Lima,
America/Los_Angeles,America/Manaus,America/Matamoros,America/Mexico_City,America/Monterrey,America/Montevideo,
America/New_York,America/Phoenix,America/Santiago,America/Sao_Paulo,America/Tijuana,America/Toronto,Asia/Amman,
Asia/Ashgabat,Asia/Baghdad,Asia/Baku,Asia/Bangkok,Asia/Beirut,Asia/Calcutta,Asia/Damascus,Asia/Dhaka,Asia
/Hong_Kong,Asia/Irkutsk,Asia/Jakarta,Asia/Jerusalem,Asia/Kabul,Asia/Karachi,Asia/Kathmandu,Asia/Kolkata,Asia
/Krasnoyarsk,Asia/Magadan,Asia/Manila,Asia/Muscat,Asia/Novosibirsk,Asia/Rangoon,Asia/Riyadh,Asia/Seoul,Asia
/Shanghai,Asia/Singapore,Asia/Taipei,Asia/Tehran,Asia/Tokyo,Asia/Ulaanbaatar,Asia/Vladivostok,Asia/Yakutsk,Asia
/Yerevan,Atlantic/Azores,Atlantic/Cape_Verde,Australia/Adelaide,Australia/Brisbane,Australia/Darwin,Australia
/Eucla,Australia/Hobart,Australia/Lord_Howe,Australia/Perth,Australia/Sydney,Brazil/DeNoronha,Brazil/East,Canada
/Newfoundland,Canada/Saskatchewan,Etc/GMT-3,Europe/Amsterdam,Europe/Athens,Europe/Berlin,Europe/Dublin,Europe
/Helsinki,Europe/Kaliningrad,Europe/London,Europe/Madrid,Europe/Moscow,Europe/Paris,Europe/Prague,Europe/Rome,
Europe/Sarajevo,Pacific/Apia,Pacific/Auckland,Pacific/Chatham,Pacific/Fiji,Pacific/Guam,Pacific/Honolulu,Pacific
/Kiritimati,Pacific/Marquesas,Pacific/Samoa,Pacific/Tongatapu,Pacific/Wake,US/Alaska,US/Central,US/East-Indiana,
US/Eastern,US/Pacific,UTC",
    "IsModifiable": true,
    "IsCollection": false
  }
},
"DBSecurityGroupMemberships": [],
"VpcSecurityGroupMemberships": []
}
},
"AllowsVpcAndNonVpcInstanceMemberships": false,
"OptionGroupArn": "arn:aws:rds:us-west-2:893547637742:og:oracle-ee-12-1-prod-clusterdb-vpc-2"
}
}
{
  "DBSubnetGroup": {
    "DBSubnetGroupName": "prod-clusterdb-vpc-2",
    "DBSubnetGroupDescription": "prod-clusterdb-vpc-2 subnet group",
    "VpcId": "vpc-0d8f1534aaab54ac6",
    "SubnetGroupStatus": "Complete",
    "Subnets": [
      {
        "SubnetIdentifier": "subnet-0f7ef67760fa393c1",
        "SubnetAvailabilityZone": {
          "Name": "us-west-2c"
        },
        "SubnetStatus": "Active"
      },
      {
        "SubnetIdentifier": "subnet-0a5dc411a66975e04",
        "SubnetAvailabilityZone": {
          "Name": "us-west-2a"
        },
        "SubnetStatus": "Active"
      },
      {
        "SubnetIdentifier": "subnet-08cc9c29050b7fc94",
        "SubnetAvailabilityZone": {
          "Name": "us-west-2b"
        },
        "SubnetStatus": "Active"
      }
    ],
    "DBSubnetGroupArn": "arn:aws:rds:us-west-2:893547637742:subgrp:prod-clusterdb-vpc-2"
  }
}
{
  "GroupId": "sg-03b55c3225c7af3d4"
}
Security Group ID: sg-03b55c3225c7af3d4
client_subnet=10.95.104.0/24
client_subnet=10.95.105.0/24
client_subnet=10.95.106.0/24
client_subnet=10.95.110.192/26
client_subnet=10.95.111.0/26
client_subnet=10.95.111.64/26
client_subnet=10.95.110.0/26
client_subnet=10.95.110.64/26

```

```
client_subnet=10.95.110.128/26
```

- Run the following command to create a RDS:

```
Usage: ./create_rds_w_cf.sh <AppEnv> <DBType> <DBName> <VPC>

where <AppEnv> should be qa, e2e, stg, sbx, prf, prod, prods, prom, prodx or prodn
      prods means small-size prod
      prodm means median-size prod
      prodx means large-size prod
      prodn means new prod (with less half of regular prod storage)
      <DBType> should be clusterdb or reportdb
      <VPC> should be vpc-1, vpc-2, etc
```

For example,

```
./create_rds_w_cf.sh prod clusterdb pitpp001 vpc-2
```

Here is the output from above command:

```
banl141bbfff3:prod kpopat$ ./create_rds_w_cf.sh prod clusterdb pitpp001 vpc-2
Create custom CMK...
kmskeyid=ba9e53f4-c6e2-4b13-a242-5b2f04b88262
{
  "StackId": "arn:aws:cloudformation:us-west-2:893547637742:stack/rds-pitpp001-20181128T062614/85c45a50-f319-11e8-944c-500c32c86c8d"
}
```

Note: It will take about 35 minutes.

- You can either use AWS console or run the following command to check the RDS creation status.

```
aws --profile sbg-psp-prod cloudformation describe-stacks --stack-name <Stack Name>
```

For example, run the command for above Perf C4 RDS creation:

```
wdhl0753fb3a0:aws-rds-master sliu$ aws --profile sbg-psp-prod cloudformation describe-stacks --stack-name "arn:aws:cloudformation:us-west-2:475968958217:stack/rds-pqbopc04-20160831T135507/34095df0-6fbd-11e6-8792-50a686fc37d2"
```

Here is the output from above command. "StackStatus": "CREATE_IN_PROGRESS" is at the end of the output. "StackStatus": "CREATE_COMPLETE" will indicate that the creation is completed. It will take about 40 minutes to complete.

```
aws --profile sbg-qbo-prod cloudformation describe-stacks --stack-name "arn:aws:cloudformation:us-west-2:475968958217:stack/rds-pqbopc04-20160831T135507/34095df0-6fbd-11e6-8792-50a686fc37d2"
```

```
{
  "Stacks": [
    {
      "StackId": "arn:aws:cloudformation:us-west-2:475968958217:stack/rds-pqbopc04-20160831T135507/34095df0-6fbd-11e6-8792-50a686fc37d2",
      "Description": "Create QBO RDS stack",
      "Parameters": [
        .....
        "StackName": "rds-pqbopc04-20160831T135507",
        "NotificationARNs": [],
        "StackStatus": "CREATE_IN_PROGRESS",
        "DisableRollback": false
      ]
    }
  ]
}
```

- Change master user (intuadmin)'s password:

```
aws --profile sbg-psp-prod rds modify-db-instance --db-instance-identifier <RDS db name> --master-user-password <new password>
```

Note: 1) Please use Initial password for production; 2) Do not double-quote <new password>

1.2 Create Goldengate Hub

- Download the following files from <https://github.com/intuit.com/payroll-services-platform/desktop-payroll-database-engineering/tree/master/goldengate-hub>. Please always redownload the files each db migration because they are keeping change for new feature or bug fix.

```
goldengate-hub.json
create_gg_hub_w_cf.sh
```

- Run the following command from your Mac to create Goldengate Hub.

Note: You should use this parameter **VolumeSize** for small production clusters or non-prod clusters, which don't need 1 TB volume. It is easy to increase volume size later on. But it is not easy to decrease it later on. We should be careful to choose volume size.

```
./create_gg_hub_w_cf.sh
Usage: ./create_gg_hub_w_cf.sh <AppEnv> <DBName> <VPC> [<VolumeSize>]
      where <AppEnv> should be qa, e2e, stg, sbx, prf or prod
            <VPC> should be vpc-1, vpc-2, etc
            [<VolumeSize>] is optional. It is EBS volume /u01 size in GB and default is 1000 GB

For example,
./create_gg_hub_w_cf.sh prf pqbopc04 vpc-1
```

Note: It will take about 13 minutes.

- Use AWS console/CloudFormation to check the progress. Proceed next step only after this is completed.

2. setup replication from primary to parallel environment

2.1 Start GoldenGate extract and pump from Primary to parallel environment

- change trail file retention to 30 days in mgr parameter file and restart mgr
- Excluded below 36 plain text columns(27 tables) and their respective _PT columns on extract. Also excluded below 3 IBOB tables from extract.

S.No.	table name	Plaint text column names
1	PSP_REPORTING_AGENT	FED_ID, FED_TAX_ID
2	PSP_TAX_CREDITS9061	FED_TAX_ID, S_S_N
3	PSP_TP401K_SIGNUP_QUEUE	FED_TAX_ID
4	PSP_PSTUB_STATE_TAX_INFO	AGENCY_ID
5	PSP_COMPANY_CONSENT	FEIN
6	PSP_EDI_PAYMENT_DETAIL	FED_TAX_ID
7	PSP_ACHENROLLMENT_DETAIL	F_E_I_N, AGENCY_ID
8	PSP_ENTITY_CHANGE	OLD_EIN, NEW_EIN
9	PSP_COMP_PMT_TEMPLATE_AGENCYID	AGENCY_TAXPAYER_ID
10	PSP_FRAUD_COMPANY	FED_TAX_ID
11	PSP_FSET_FILING_DETAIL	FED_TAX_ID, AGENCY_ID
12	PSP_RAFENROLLMENT_DETAIL	FED_TAXID
13	PSP_EFTPS_ENROLLMENT_DETAIL	FED_TAX_ID
14	PSP_FRAUD_BANK_ACCOUNT	ACCOUNT_NUMBER
15	PSP_COMPANYAGENCY_PMTTEMPLATE	AGENCY_TAXPAYER_ID
16	PSP_FRAUD_EVENT	COMPANY_EIN
17	PSP_PAYEE	ACCOUNT_NUMBER, TAX_ID
18	PSP_CONTACT	SOCIAL_SECURITY_NUMBER, DATE_OF_BIRTH
19	PSP_ENTITLEMENT_UNIT	FED_TAX_ID
20	PSP_QBDT_PAYROLL_ITEM_INFO	AGENCY_ID
21	PSP_COMPANY	FED_TAX_ID
22	PSP_EFTPS_PAYMENT_DETAIL	FED_TAX_ID
23	PSP_BANK_ACCOUNT	ACCOUNT_NUMBER
24	PSP_EMPLOYEE	TAX_ID, BIRTH_DATE
25	PSP_MONEY_MOVEMENT_TRANSACTION	AGENCY_TAXPAYER_ID
26	PSP_ENTRY_DETAIL_RECORD	RECORD_DATA, TXP_RECORD_DATA
27	PSP_DEPOSIT_FREQUENCY_FILE_REC	E_I_N

3 IBOB tables:

S.No.	Table Name
1	PSP_SOURCE_SYSTEM_TRANSMISSION
2	PSP_QBDT_REQUEST_INFO
3	PSP_SAP_METHOD_CALL

- start extract and pump on primary
- Note down current/latest trail file sequence number for later verification

```
GGSCI (pprdpspdb901.ie.intuit.net) 13> info all
```

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	STOPPED	EIB0BPD2	00:00:02	49:35:56
EXTRACT	RUNNING	EPSPLPD2	00:00:04	00:00:00
EXTRACT	RUNNING	PPSPLPD1	00:00:00	00:00:06

```
GGSCI (pprdpspdb901.ie.intuit.net) 14> sh ls -ltr ./dirdat/psplpd02/tr*|tail
```

```
-rw-r----- 1 oracle dba 499999751 Nov 21 14:32 ./dirdat/psplpd02/tr000011776
-rw-r----- 1 oracle dba 499999849 Nov 21 15:04 ./dirdat/psplpd02/tr000011777
-rw-r----- 1 oracle dba 499999437 Nov 21 15:37 ./dirdat/psplpd02/tr000011778
-rw-r----- 1 oracle dba 499999338 Nov 21 16:34 ./dirdat/psplpd02/tr000011779
-rw-r----- 1 oracle dba 499991973 Nov 21 16:55 ./dirdat/psplpd02/tr000011780
-rw-r----- 1 oracle dba 499992781 Nov 21 17:24 ./dirdat/psplpd02/tr000011781
-rw-r----- 1 oracle dba 499999693 Nov 21 17:59 ./dirdat/psplpd02/tr000011782
-rw-r----- 1 oracle dba 499999954 Nov 21 20:57 ./dirdat/psplpd02/tr000011783
-rw-r----- 1 oracle dba 227653325 Nov 22 01:33 ./dirdat/psplpd02/tr000011784
-rw-r----- 1 oracle dba 1370489 Nov 22 01:36 ./dirdat/psplpd02/tr000011785
```

- Ask storage team to take latest snapshot backup and start restore on parallel environment. note incomplete recovery SCN

```
RESETLOGS after incomplete recovery UNTIL CHANGE 8385767210214 time 11/28/2018 01:54:25
```

2.2 start replicate

On Target DB server:

- set up supplemental log

```
SELECT SUPPLEMENTAL_LOG_DATA_MIN, FORCE_LOGGING FROM V$DATABASE;

ALTER DATABASE ADD SUPPLEMENTAL LOG DATA;
ALTER DATABASE FORCE LOGGING;
ALTER SYSTEM SWITCH LOGFILE;

SELECT SUPPLEMENTAL_LOG_DATA_MIN, FORCE_LOGGING FROM V\ $DATABASE;
```

- Grant permission to GGT

```
grant create session to GGT;
grant DBA to GGT;
exec dbms_goldengate_auth.grant_admin_privilege ('GGT');
grant exempt access policy to GGT;

GRANT SELECT ANY DICTIONARY TO GGS;
```

- Create credentials and checkpoint. You need to replace <RDS SID> and <password> before running the command.

```
obey ./diroby/dblogin_t.oby
add checkpointtable
```

- Add schematrandata

```
obey ./diroby/dblogin_s.oby
delete schematrandata pspadm
add schematrandata pspadm
```

3. export backup on parallel environment

3.1 Nullify plain text columns on target

- execute below script to nullify plain text columns – **completed**

- drop TDE wallet
- create restore point

```
SQL> SELECT NAME, SCN, TIME, DATABASE_INCARNATION#, GUARANTEE_FLASHBACK_DATABASE, STORAGE_SIZE FROM
V$RESTORE_POINT;
NAME                                SCN TIME
DATABASE_INCARNATION# GUA          STORAGE_SIZE
-----
NOV29_0715                        8385771428212 29-NOV-18 07.15.58.000000000 AM
2 YES                             10415562752
```

3.2 start export backup PSPADM schema by excluding IBOB tables

- Take export backup of PSPADM schema by excluding PSP_SAP_METHOD_CALL, PSP_SOURCE_SYSTEM_TRANSMISSION, PSP_QBDT_R EQUEST_INFO tables
- Export data from primary db server (QDC or LVDC), based on primary mode. Use DOP 32 and dump file size limit 150 G
- Go to the directory

```
cd ~/AWS/$ORACLE_SID/scripts
```

- Create directories in the source db

```
sqlplus / as sysdba @A_create_directories.sql $ORACLE_SID
```

- Create the database link to the target db in the source db

```
sqlplus / as sysdba @B_create_db_links.sql
```

- Export the entire metadata

```
nohup ./C_exp_metadata.sh &
```

Verify the export log ../logs/metadata.log

```
grep Job ../logs/metadata.log Job "SYS"."SYS_EXPORT_FULL_01" successfully completed at Thu Oct 13 14:15:33 2016
elapsed 0 00:29:50
```

- Export the data from PSPADM schema. It will temporally increase undo_retention to 6 hours (21600 seconds). **We may need to think about undo_retention for PSP.**
- export command would look like as below .

```
expdp userid=\"/ as sysdba\" directory=WORK_DIR dumpfile=exp_data%U.dmp filesize=150g parallel=32
schemas=PSPADM EXCLUDE=TABLE:\"IN (\"PSPADM.PSP_SOURCE_SYSTEM_TRANSMISSION\", \"PSPADM.PSP_QBDT_REQUEST_INFO\",
\"PSPADM.PSP_SAP_METHOD_CALL\")\" content=data_only logfile=WORK_LOG:exp_data.log
```

```
nohup ./D_exp_data.sh &
```

Verify the export log ../logs/exp_data.log

```
grep Job ../logs/exp_data.log
Job "SYS"."SYS_EXPORT_SCHEMA_01" successfully completed at Thu Oct 13 14:54:21 2016 elapsed 0 00:29:57
```

- Check # of dump files. If there are more than 16 dump files, create additional scripts E_copy_dump_files_to_target_p[5-?].sql and add the script names to E_copy_dump_files_to_target_all.sh
- Transfer dump files to RDS

```
./E_copy_dump_files_to_target_all.sh
```

Note:

1. Check the log file E_copy_dump_files_to_target_p[0-4].lst for file transferring completion
2. You can run the following command on the goldengate hub server to check the progress

```
./run_on_target.sh chk_imp_file.sql
```

Verify if the transferring was successful

```
oracle@pprdqbodb801.ie.intuit.net:/home/oracle/AWS/qboqpd34/scripts->egrep "successfully completed|ORA-"
E_copy_dump_files_to_target_p*.lst
E_copy_dump_files_to_target_p0.lst:PL/SQL procedure successfully completed.
E_copy_dump_files_to_target_p1.lst:PL/SQL procedure successfully completed.
E_copy_dump_files_to_target_p2.lst:PL/SQL procedure successfully completed.
E_copy_dump_files_to_target_p3.lst:PL/SQL procedure successfully completed.
E_copy_dump_files_to_target_p4.lst:PL/SQL procedure successfully completed.
```

Note: You may get the errors like below if # of dump files is less than 16. You can ignore the errors in this case.

```
E_copy_dump_files_to_target_p4.lst:ORA-19505: failed to identify file
E_copy_dump_files_to_target_p4.lst:ORA-27037: unable to obtain file status
E_copy_dump_files_to_target_p4.lst:ORA-06512: at "SYS.DBMS_FILE_TRANSFER", line 60
E_copy_dump_files_to_target_p4.lst:ORA-06512: at "SYS.DBMS_FILE_TRANSFER", line 168
E_copy_dump_files_to_target_p4.lst:ORA-06512: at line 3
```

You may get the error like "ORA-03150: end-of-file on communication channel for database link". You need to delete the incomplete dumps file on the RDS by running the following command on the goldengate hub server

```
./run_on_target.sh clean_imp_file_p.sql <file name> or ./run_on_target.sh clean_imp_file_all.sql
```

4. setup replication from primary to AWS

4.1 Create Goldengate Hub

4.2 Setup the migration environment on Goldengate Hub

- Create the directories on Goldengate hub server
- Download the migration target script from S3 and untar it.
- Change the scripts login_target.sh and run_on_target.sh with the correct target RDS connection string and credential. Verify by login

4.3 Import data to the target RDS db

- Do below by login to GG hub
- Recreate redo logs

```
./run_on_target.sh 00_add_redos.sql
```

Verify if this is successful

```
grep ORA- 00_add_redos.lst
```

- Create tablespaces.
 - Please review and change script 01_create_tablespaces.sql for PSP tablespaces needs to be created and change the initial and incremental size properly before running it.
 - Please don't run it when the dump file transferring job hasn't completed; otherwise likely that job will failed with the error like "ORA-03150: end-of-file on communication channel for database link".

```
nohup ./run_on_target.sh 01_create_tablespaces.sql &
```

Verify if this is successful

```
grep ORA- 01_create_tablespaces.lst
```

- Create schema owners

```
./run_on_target.sh 02_create_schema_owners.sql
```

- Verify if this is successful

```
grep ORA- 02_create_schema_owners.lst
```

- Import metadata

```
nohup ./run_on_target.sh 03_import_metadata.sql &
```

- Run the following command to watch the job completion

```
./watch_db_job.sh run_on_target.sh read_file_p.sql import_metadata.log  
At Sat Sep 3 11:11:57 PDT 2016: return=1
```

Verify if there are any unexpected error

```
[oracle@pprdc34uw214g238195 scripts]$ grep ORA- import_metadata.log |egrep -v "ORA-31685|ORA-31684|ORA-31684|ORA-02380|ORA-01917|ORA-39083|ORA-20997|ORA-01932|ORA-01924|ORA-01918|ORA-01950|ORA-39151|ORA-31625|ORA-22930|ORA-29371|ORA-01435|ORA-01435|ORA-39112|ORA-06564|SYS|ORA-06550|ORA-20900|ORA-00955|ORA-39342|ORA-39325|ORA-04052|ORA-04042|ORA-12154|ORA-00604"
```

- Drop all PSPADM constraints

```
./run_on_target.sh 04_drop_constraints.sql
```

Verify if there are any error

```
grep ORA- 04_drop_constraints.lst
```

- Disable all PSPADM triggers. **Needs review.**

```
./run_on_target.sh 05_disable_triggers.sql
```

Verify if there are any error

```
grep ORA- 05_disable_triggers.lst
```

- Import PSPADM data

```
nohup ./run_on_target.sh 06_import_data.sql &
```

Run the following command to watch the job completion

```
./watch_db_job.sh run_on_target.sh read_file_p.sql import_data.log
At Sat Sep 3 11:24:02 PDT 2016: return=0
At Sat Sep 3 11:29:08 PDT 2016: return=0
...
At Sat Sep 3 15:28:44 PDT 2016: return=1
```

You can check the import progress with the following command too

```
[oracle@pprdc34uw214g238195 scripts]$ ./run_on_target.sh chk_datapump_status.sql INTUADMIN.IMPORT_DATA |grep
Done
16:12:02 79      dbms_output.put_line(RPAD('% Done          : ' || js.percent_done || ' % <----- *****',
col_pad_len ) || RPAD('Restart Count   : ' || js.restart_count,col_pad_len ) );
16:12:02 163      || ' , ' || LPAD(wsl(ind).percent_done,2) || '% Done'
16:12:02 164      --|| ' , ' || LPAD(percent_done,6) || '% Done'
% Done          : 47 % <----- ***** Restart Count   : 0
Worker 1 : DW00 , 0% Done , Deg 0 , WORK WAIT , Completed Objects 0/0 , 0 K
...
```

Check errors in import_data.log

```
cat import_data.log |grep Job Job "INTUADMIN"."IMPORT_DATA" successfully completed at Fri Oct 14 18:56:46
2016 cat import_data.log |grep ORA- (nothing returned)
```

- Collect table statistics. **Needs review.**

```
nohup ./run_on_target.sh 07_collect_stats.sql &
```

After finish, verify if there are any errors

```
grep ORA- 07_collect_stats.lst
```

Note: We skipped above collect_stats step as we exported stats from source/parallel env and imported to AWS RDS.

```
1. SQL> exec dbms_stats.export_schema_stats( 'PSPADM', 'STATS_11DEC' );  
  
PL/SQL procedure successfully completed.  
  
2. export table STATS_11DEC and copy dump file to AWS RDS  
  
3. import table STATS_11DEC to AWS RDS  
  
4. SQL> exec dbms_stats.import_schema_stats('PSPADM','STATS_11DEC');  
  
PL/SQL procedure successfully completed.  
  
Elapsed: 00:06:42.23
```

- Import indexes and constraints

```
nohup ./run_on_target.sh 08_import_index_a_const.sql &
```

Run the following command to watch the job completion

```
./watch_db_job.sh run_on_target.sh read_file_p.sql import_index_a_const.log  
At Sat Sep 3 15:42:40 PDT 2016: return=0  
At Sat Sep 3 15:47:46 PDT 2016: return=0  
...  
At Sun Sep 4 02:45:20 PDT 2016: return=1
```

Check errors in import_index_a_const.log

```
[oracle@pprdc34uw214g238195 scripts]$ cat import_index_a_const.log |grep Job  
Job "INTUADMIN"."IMPORT_INDEX_A_CONST" successfully completed at Sat Oct 15
```

- Enable triggers. **Needs review.**


```
./run_on_target.sh 09_enable_triggers.sql
```

Verify if there are any errors

```
grep ORA- 09_enable_triggers.lst
```

- Lock all app user accounts. **Needs review of APP users to be locked.**

```
./run_on_target.sh lock_app_users.sql
```

Verify if there are any errors

```
grep ORA- LOCK_APP_USERS.lst
```

- Compile all invalidated objects

```
./run_on_target.sh 13_compile_objects.sql
```

At the end of the output, you will see the remaining invalidated objects:

```
SQL> select owner, object_type, object_name from dba_objects where status <> 'VALID';
```

4.3.1 Sanity check

- Run the following SQLs on both source and target and then compare the results. **Need to change for PSPADM owned objects.**

```
select owner, count(*) from dba_tables where owner in ('PSPADM') group by owner order by 1;
select index_type, count(*) from dba_indexes where owner in ('PSPADM') group by index_type order by 1;
select constraint_type, count(*) from dba_constraints where owner in ('PSPADM') group by constraint_type order by 1;
select status, count(*) from dba_triggers where owner in ('PSPADM') group by status;
```

Output would look like as below :

On source/parallel env:

```
SQL> select owner, count(*) from dba_tables where owner in ('PSPADM') group by owner order by 1;
```

OWNER	COUNT(*)
PSPADM	365

```
SQL> select index_type, count(*) from dba_indexes where owner in ('PSPADM') group by index_type order by 1;
```

INDEX_TYPE	COUNT(*)
FUNCTION-BASED NORMAL	6
LOB	24
NORMAL	779

```
SQL> select constraint_type, count(*) from dba_constraints where owner in ('PSPADM') group by constraint_type order by 1;
```

C	COUNT(*)
C	1438
P	291
R	416
U	2

```
SQL> select status, count(*) from dba_triggers where owner in ('PSPADM') group by status;
```

STATUS	COUNT(*)
ENABLED	23

On Target/AWS:

```
SQL> select owner, count(*) from dba_tables where owner in ('PSPADM') group by owner order by 1;
```

OWNER	COUNT(*)
PSPADM	365

```
SQL> select index_type, count(*) from dba_indexes where owner in ('PSPADM') group by index_type order by 1;
```

INDEX_TYPE	COUNT(*)
FUNCTION-BASED NORMAL	6
LOB	24
NORMAL	779

```
SQL> select constraint_type, count(*) from dba_constraints where owner in ('PSPADM') group by constraint_type order by 1;
```

C	COUNT(*)
C	1438
P	291
R	416
U	2

```
SQL> select status, count(*) from dba_triggers where owner in ('PSPADM') group by status;
```

STATUS	COUNT(*)
--------	----------

4.4 Final step for Goldengate replication from IHP to AWS

On Godengate Hub

- Login RDS

```
~/AWS/$RDS_SID/scripts/login_target.sh
```

- Setup supplemental log

```
SELECT SUPPLEMENTAL_LOG_DATA_MIN, FORCE_LOGGING FROM V$DATABASE;  
exec rdsadmin.rdsadmin_util.alter_supplemental_logging('ADD');  
exec rdsadmin.rdsadmin_util.force_logging();  
exec rdsadmin.rdsadmin_util.switch_logfile;  
exec rdsadmin.rdsadmin_util.set_configuration('archive log retention hours',72);  
commit;
```

- Grant permission to GGT and GGS

```
grant create session to GGT;  
grant DBA to GGT;  
exec dbms_goldengate_auth.grant_admin_privilege ('GGS');  
exec dbms_goldengate_auth.grant_admin_privilege ('GGT');  
grant exempt access policy to GGT;  
GRANT SELECT ANY DICTIONARY TO GGS;
```

- Create credentials and checkpoint. You need to replace <RDS SID> and <password> before running the command.

```

GGSCI (ip-10-86-14-162.us-west-2.compute.internal) 1> add CREDENTIALSTORE Credential store created in ./dircrd
/.
GGSCI (ip-10-86-14-162.us-west-2.compute.internal) 2> ALTER CREDENTIALSTORE ADD USER ggs@<RDS SID> PASSWORD
<password> ALIAS ggsources
Credential store in ./dircrd/ altered.
GGSCI (ip-10-86-14-162.us-west-2.compute.internal) 3> ALTER CREDENTIALSTORE ADD USER ggt@<RDS SID> PASSWORD
<password> ALIAS ggtarget
Credential store in ./dircrd/ altered.
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com) 11> obey ./diroby/dblogin_s.oby
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com) 13> obey ./diroby/dblogin_t.oby
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com) 14> add checkpointtable
No checkpoint table specified.
Using GLOBALS specification (GGT.CHECKPOINT)... ERROR: Failed creating checkpoint table GGT.CHECKPOINT.
OCI Error ORA-00955: name is already used by an existing object (status = 955), SQL <CREATE TABLE GGT.
CHECKPOINT_lox ( group_name VARCHAR2(8) NOT NULL, group_key NUMBER(19) NOT NULL, log_cmplt_csn VARCHAR2(128)
NOT NULL, log_cmplt_xids_seq NUMBER(5) NOT NULL, log_cmplt_xids VARCHAR2(2000) NOT NULL, PRIMARY KEY
(group_name, group_key, log_cmplt_csn, log_cmplt_xids_seq))>. Note: Ignore the about error
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com) 15> exit

```

- Add schematrandata

```

GGSCI (pcuw216g238166.us-west-2.compute.internal) 1> obey ./diroby/dblogin_s.oby
GGSCI (pcuw216g238166.us-west-2.compute.internal) 2> dblogin USERIDALIAS ggsources
Successfully logged into database.
GGSCI (pcuw216g238166.us-west-2.compute.internal as ggs@QB0PP092) 3> add schematrandata PSPADM

```

- Create rds_sequence.sql and run it on AWS db

```

1. Go to $GG_HOME

2. Create the script rds_sequence.sql
sed '8,25d' sequence.sql > rds_sequence.sql

3. Login to the RDS and run the script

~/AWS/${RDS_SID}/scripts/login_target.sh
SQL> @rds_sequence.sql
Please enter the name of a schema for the GoldenGate database objects:
ggs
Setting schema name to GGS

UPDATE_SEQUENCE STATUS:

Line/pos    Error
-----
No errors   No errors

GETSEQFLUSH

Line/pos    Error
-----
No errors   No errors

SEQTRACE

Line/pos    Error
-----
No errors   No errors

REPLICATE_SEQUENCE STATUS:

Line/pos    Error
-----
No errors   No errors

STATUS OF SEQUENCE SUPPORT
-----
SUCCESSFUL installation of Oracle Sequence Replication support
SQL>

```

- Grant permission to GGT

```

GRANT EXECUTE ON GGS.updateSequence to GGT;
GRANT EXECUTE ON GGS.replicateSequence to GGT;

```

- Setup and start the IHP to AWS replicat process on Goldengate hub. [Need to change below output as per PSP env.](#)

```

GGSCI (ggpitpp001.sbg-ppsp-prod.a.intuit.com) 2> obey ./diroby/create_rpsplpd2.oby

GGSCI (ggpitpp001.sbg-ppsp-prod.a.intuit.com) 3> dblogin USERIDALIAS ggtarget

Successfully logged into database.

GGSCI (ggpitpp001.sbg-ppsp-prod.a.intuit.com as ggt@PITPP001) 4> add replicat rpsplpd2, INTEGRATED, EXTTRAIL .
./dirdat/psplpd02/tr

REPLICAT (Integrated) added.

GGSCI (ggpitpp001.sbg-ppsp-prod.a.intuit.com as ggt@PITPP001) 5> info rpsplpd2

REPLICAT   RPSPLPD2   Initialized   2018-12-13 23:17   Status STOPPED
INTEGRATED
Checkpoint Lag      00:00:00 (updated 00:00:03 ago)
Log Read Checkpoint File ./dirdat/psplpd02/tr0000000000
First Record   RBA 0

GGSCI (ggpitpp001.sbg-ppsp-prod.a.intuit.com as ggt@PITPP001) 6> sh ls -ltr ./dirdat/psplpd02/tr0000000000

-rw-r-----. 1 oracle oinstall 499999412 Dec 13 22:37 ./dirdat/psplpd02/tr0000000000

GGSCI (ggpitpp001.sbg-ppsp-prod.a.intuit.com as ggt@PITPP001) 7> dblogin USERIDALIAS ggtarget
Successfully logged into database.

GGSCI (ggpitpp001.sbg-ppsp-prod.a.intuit.com as ggt@PITPP001) 8> register replicat rpsplpd2 database

2018-12-13 23:18:38 INFO      OGG-02528 REPLICAT RPSPLPD2 successfully registered with database as inbound
server OGG$RPSPLPD2.

-----> start RPSPLPD2 ATCSN <SCN from data export step>

GGSCI (ggpitpp001.sbg-ppsp-prod.a.intuit.com as ggt@PITPP001) 8> start RPSPLPD2 ATCSN 8385767210214

Sending START request to MANAGER ...
REPLICAT RPSPLPD2 starting

```

- we had to disable below 2 constraints on 2 tables for which PSP_SOURCE_SYSTEM_TRANSMISSION is parent table

```

SQL> alter table pspadm.PSP_TRANSMISSION_PAYROLL_RUN disable constraint PSP_TRANSMISSIONPAYROLLRUN_FK1;

Table altered.

SQL> alter table pspadm.PSP_QBDT_UNPROCESSED_REQUEST disable constraint PSP_QBDT_UNPROCESSED_REQUE_FK1;

Table altered.

```

- **On the source side**, run the following GG commands to flush sequences. **Need to change below output as per PSP env.**

```
# If primary db is in QDC:
GGSCI (pprfqbodb802.ie.intuit.net) 3> obey ./diroby/dblogin_s.oby
# If primary db is in LVDC:
GGSCI (pprfqbodb802.ie.intuit.net) 3> obey ./diroby/dblogin_s_remote.oby

GGSCI (pprfqbodb802.ie.intuit.net) 4> dblogin USERIDALIAS ggs source

Successfully logged into database.

GGSCI (pprfqbodb802.ie.intuit.net as ggs@pqboc04) 5> flush sequence QBO.*
Successfully flushed 7 sequence(s) QBO.*.

GGSCI (pprfqbodb802.ie.intuit.net as ggs@pqboc04) 6> flush sequence QBO_DATA.*
Successfully flushed 19 sequence(s) QBO_DATA.*.
```

- Upload the credential wallet to S3 by running the following commands on GG Hub. **Need to review and change.**

```
. /l/orcl
region="us-west-2"
bucket_name="iss-094918931292-us-west-2"
kms_id="277fea6c-5f01-42e7-abf0-43954479f332"
cluster_env="prod"
cluster_num=`echo $RDS_SID|cut -c7-8`
/usr/local/bin/secrets put --region $region --s3-bucket $bucket_name --kms-cmk-id $kms_id --secret-names secrets
/goldengate/${cluster_env}/c${cluster_num}/dircrd/cwallet.sso --secret-value file://$GG_HOME/dircrd/cwallet.sso
```

This is for prod.

The output will like below: **Need to change below output as per PSP env.**

```
log4j:WARN Cannot create KAOS log file from /var/log/secrets-clidirectory. Please check if you have right
permissions. [ { "secretName": "secrets/goldengate/prod/c60/dircrd/cwallet.sso", "secretVersion":
"8K5bvKG7oanQF.oBWLwOw7x6jyT9veTu" } ]
```

- Wait until there is no replication lag showed up from "info all" command
- Create the Goldengate heartbeat table on IHP

```
# If primary db is in QDC: cd /u01/ogg/scripts sqlplus / as sysdba @monitor_gg_sync_table_ihp.sql # If
primary db is in LVDC vi monitor_gg_sync_table_ihp.sql - changing "connect ops_user/&1" to "connect ops_user
/&1@<Primary db name>" sqlplus sys@<primary db name> as sysdba @monitor_gg_sync_table_ihp.sql
```

- On AWS, verify GG replication

```
SQL> alter session set nls_date_format='yyyy-mm-dd hh24:mi:ss'; Session altered. SQL> select * from pspadm.
gg_heartbeat; SOURCE LAST_UPDATE ----- IHP 2017-02-11 11:52:00 Make sure
that the record IHP is close to current time (the difference should not be great than 1 minute)
```

- Run the following scrips to verify sequence replication. **Need to review and change.**

```
On Source: cd ~/AWS/$ORACLE_SID/scripts sqlplus / as sysdba @I_chk_sequences.sql On Target: cd ~/AWS/$RDS_SID
/scripts ./run_on_target.sh 15_chk_sequences.sql
```

The output will looks like below and #s on target will be always not smaller than #s on source

4.5 Setup Veridata

- Set up wiki and provide a link here to setup veridata between IHP and AWS.

4.6 Setup Goldengate replication from AWS to IHP

1. **Perform this step only after Veridata comparison shows that the databases are sync.**
 2. **As soon as you start to perform this step, you should not perform any data fix or recreating schema or tables in the RDS because it will be replicated to the DB in IHP. Likely you have to clean up the AWS to IHP replication before you perform these tasks.**
- On AWS, create extract and pump (to LVDC) . **Need to change below output as per PSP env.**


```

GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com) 1> obey ./diroby/create_eqbpp034.oby
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com) 2> dblogin USERIDALIAS ggs@QB0PP034 Successfully
logged into database.
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com as ggs@QB0PP034) 3> add extract eqbpp034,
INTEGRATED TRANLOG, BEGIN NOW EXTRACT added.
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com as ggs@QB0PP034) 4> add exttrail ./dirdat
/qbopp034/tr extract eqbpp034, MEGABYTES 500 EXTTRAIL added.
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com as ggs@QB0PP034) 5> info all
Program Status Group Lag at Chkpt Time Since Chkpt
MANAGER RUNNING
EXTRACT STOPPED EQBPP034 00:00:00 00:00:16
REPLICAT RUNNING RQBQPD34 00:00:02 00:00:04
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com as ggs@QB0PP034) 6> register extract EQBPP034
database
Extract EQBPP034 successfully registered with database at SCN 8451888176186.
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com) 1> obey ./diroby/create_pqbqpd34.oby
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com) 2> add extract pqbqpd34, EXTTRAILSOURCE ./dirdat
/qbopp034/tr EXTRACT added.
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com) 3> add rmttrail ./dirdat/qbopp034/tr, extract
pqbqpd34, MEGABYTES 500 RMTTRAIL added.
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com) 4>
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com as ggs@QB0PP034) 5> info all
Program Status Group Lag at Chkpt Time Since Chkpt
MANAGER RUNNING
EXTRACT STOPPED EQBPP034 00:00:00 00:00:16
EXTRACT STOPPED PQBQPD34 00:00:00 00:11:21
REPLICAT RUNNING RQBQPD34 00:00:02 00:00:04

```

- On AWS, start extract and pump . [Need to change below output as per PSP env.](#)

```

GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com as ggs@QB0PP034) 5> info all
Program Status Group Lag at Chkpt Time Since Chkpt
MANAGER RUNNING
EXTRACT STOPPED EQBPP034 00:00:00 00:00:16
EXTRACT STOPPED PQBQPD34 00:00:00 00:11:21
REPLICAT RUNNING RQBQPD34 00:00:02 00:00:04
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com as ggs@QB0PP034) 7> start EQBPP034
Sending START request to MANAGER ... EXTRACT EQBPP034 starting
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com as ggs@QB0PP034) 20> info all
Program Status Group Lag at Chkpt Time Since Chkpt
MANAGER RUNNING
EXTRACT RUNNING EQBPP034 00:04:34 00:00:08
EXTRACT STOPPED PQBQPD34 00:00:00 00:16:30
REPLICAT RUNNING RQBQPD34 00:00:04 00:00:03
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com as ggs@QB0PP034) 21> sh ls -l ./dirdat/qbopp034
/tr*
-rw-r----- 1 oracle oinstall 1525 Nov 10 09:43 ./dirdat/qbopp034/tr000000
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com as ggs@QB0PP034) 24> start PQBQPD34
Sending START request to MANAGER ...
EXTRACT PQBQPD34 starting
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com as ggs@QB0PP034) 26> info all
Program Status Group Lag at Chkpt Time Since Chkpt
MANAGER RUNNING
EXTRACT RUNNING EQBPP034 00:00:02 00:00:08
EXTRACT RUNNING PQBQPD34 00:00:00 00:00:07
REPLICAT RUNNING RQBQPD34 00:00:02 00:00:04

```

- On IHP, verify the trail file from AWS. [Need to change below output as per PSP env.](#)

```

oracle@pprdqbodb801.ie.intuit.net:/u01/ogg/qboppd34/12.1.2.1.6->ls -l $GG_HOME/dirdat/qbopp034/ total 0 -rw-
r----- 1 oracle dba 0 Nov 10 09:45 tr000000

```

- On IHP, create replicat

If primary db is in LVDC, change ggtarget in the script create_rqbpp0xx.oby to ggtarget_remote first . **Need to change below output as per PSP env.**

```
GGSCI (pprdqbodb80l.ie.intuit.net as ggt@qboqpd34) 4> obey ./diroby/create_rqbpp034.oby GGSCI (pprdqbodb80l.ie.intuit.net as ggt@qboqpd34) 5> dblogin USERIDALIAS ggtarget Successfully logged into database. GGSCI (pprdqbodb80l.ie.intuit.net as ggt@qboqpd34) 6> add replicat rqbpp034, INTEGRATED, EXTTRAIL ./dirdat/qbopp034/tr REPLICAT (Integrated) added. # NOTE: If primary db is in LVDC, run "obey ./diroby/dblogin_t_remote.oby GGSCI (pprdqbodb80l.ie.intuit.net as ggt@qboqpd34) 7> obey ./diroby/dblogin_t.oby GGSCI (pprdqbodb80l.ie.intuit.net as ggt@qboqpd34) 8> dblogin USERIDALIAS ggtarget Successfully logged into database. GGSCI (pprdqbodb80l.ie.intuit.net as ggt@qboqpd34) 10> register replicat RQBPP034 database 2016-11-09 14:30:13 INFO OGG-02528 REPLICAT RQBPP034 successfully registered with database as inbound server OGG$RQBPP034.
```

- On IHP, start replicat. **Need to change below output as per PSP env.**

```
GGSCI (pprdqbodb80l.ie.intuit.net) 4> start RQBPP034 Sending START request to MANAGER ... REPLICAT RQBPP034 starting GGSCI (pprdqbodb80l.ie.intuit.net) 9> info all Program Status Group Lag at Chkpt Time Since Chkpt MANAGER RUNNING EXTRACT RUNNING EQBOPD34 00:00:02 00:00:09 EXTRACT RUNNING EQBOPD34 00:00:02 00:00:06 EXTRACT RUNNING PQBOPD34 00:00:00 00:00:01 EXTRACT RUNNING PQBPP034 00:00:00 00:00:01 REPLICAT RUNNING RQBPP034 00:00:00 00:00:02
```

- On AWS, run the following GG commands to flush sequences. **Need to change below output as per PSP env.**

```
GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com) 1> obey ./diroby/dblogin_s.oby GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com) 2> dblogin USERIDALIAS ggsources Successfully logged into database. GGSCI (pprdc34uw214g238195.us-west-2.sbg-qbo-prod.a.intuit.com as ggs@QBOP034) 3> flush sequence QB0.* Successfully flushed 6 sequence(s) QB0.*.
```

- On AWS, setup GG heartbeat monitor script

```
[oracle@pprdc34uw214g238195 scripts]$ cd /u01/ogg/scripts/ [oracle@pprdc34uw214g238195 scripts]$ ~/AWS/$RDS_SID/scripts/login_target.sh SQL> @monitor_gg_sync_table_aws.sql ... Enter ops_user password and db connection string Enter value for 1: <password> Enter value for 2: <RDS_SID>
```

- On IHP, verify GG replication

```
SQL> alter session set nls_date_format='yyyy-mm-dd hh24:mi:ss'; Session altered. SQL> select * from qbo.gg_heartbeat; SOURCE LAST_UPDATE ----- IHP 2016-11-10 11:09:58 AWS 2016-11-10 11:10:01 Make sure that the record AWS is close to current time (the difference should not be great than 1 minute)
```

- Verify sequence bi-directory replication. **Need to change below output as per PSP env.**

1. On IHP, create test sequence if it does not exists: SQL> create sequence qbo_data.test; 2. Run the following SQL on IHP and AWS to confirm the sequence exists SQL> select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER = 'QBO_DATA' and SEQUENCE_NAME = 'TEST'; 3. Test the sequence a) Run it on IHP: SQL> select qbo_data.test.nextval from dual; NEXTVAL ----- 1 b) Wait 10 seconds. Run it on AWS SQL> select qbo_data.test.nextval from dual; NEXTVAL ----- 22 Note: the # here should be higher than one in step above. If not, replica this step until this is true. If this is not case after trying for a couple of minutes c) Wait 10 seconds. Run it on IHP SQL> select qbo_data.test.nextval from dual; NEXTVAL ----- 43 Note: the # here should be higher than one in step above. If not, replica this step until this is true. If this is not case after trying for a couple of minutes

4.7 Setup RDS and GG monitoring

- On AWS, add new extract and pump to monitor_qbopp034.cfg. **Need to change below output as per PSP env.**

```
[oracle@pprdc34uw214g238195 scripts]$ cd /u01/ogg/scripts [oracle@pprdc34uw214g238195 scripts]$ cp
monitor_pqbopc04.cfg monitor_$RDS_SID.cfg Modify the new file based on which processes need to be monitored.
[oracle@pprdc34uw214g238195 scripts]$ cat monitor_qbopp034.cfg MANAGER,1 EQBPP034,2 EIAPP034,0 PQBQPD34,2
PIAPP034,0 RQBQPD34,2 NOTE: The monitor configuration file (monitor_qboqpd34.cfg), has one entry for the
manager and one entry for each of the processes (Extract, pump or replicat). The number after each entry
separated by a comma, is simply a flag: 0 = do not monitor this process 1 = monitor this process 2 = monitor
this process and additionally report on the lag and checkpoint
```

- On AWS, create cron job entry. **Need to change below output as per PSP env. Need to change crontab as well accordingly.**

```
crontab crontab.txt
```

And then change the frequency from 5 min to 1 hour

- On IHP, add new replicat to monitor_qboqpd34.cfg. **Need to change below output as per PSP env.**

```
oracle@pprdqbodb801.ie.intuit.net:/u01/ogg/qboqpd34/12.1.2.1.6->cd /u01/ogg/scripts/ oracle@pprdqbodb801.ie.
intuit.net:/scripts/ogg->cat monitor_qboqpd34.cfg MANAGER,1 EQBOPD34,2 EQBQPD34,2 PQBOPD34,2 PQBPP034,2
RQBPP034,2
```

- On IHP, create the file called login under /u01/ogg/scripts/

```
cd /u01/ogg/scripts vi login cat login ops_user/"<real password here>" chmod 700 login
```

- On IHP, make sure to have the following crontab entries

```
##Goldengate monitoring## 0 * * * * cd /u01/ogg/scripts; ./monitor_ggserr.sh ihp qboqpd34 1>/dev/null 2>&1 58
23 * * * * cd /u01/ogg/scripts; ./rotate_ggserr.sh ihp qboqpd34 1>/dev/null 2>&1 0 * * * * cd /u01/ogg/scripts; .
/monitor_gg_sync.sh 10 ihp qboqpd34 1>/dev/null 2>&1
```